

1 Problem Description

$\log_2(x) = \log_3(y)$ passes through (1,1) and (8,27).

A is the area between the curve and x axis between $x=1$ and $x=8$

B is the area between the curve and y axis between $y=1$ and $y=27$

Find $|A - B|$

2 Problem Solution

$$y = 3^{\log_2(x)}$$
$$\int_1^8 3^{\log_2(x)} dx$$

Sub into calculator and get: $A = 83.17335$

$$x = 2^{\log_3(y)}$$
$$\int_1^2 2^{\log_3(y)} dy$$

Sub into calculator and get: $B = 131.8266$

$$|A - B| = 48.65329$$